Sheet 45 Tue. 2 March 2027 Affine functions: rate of change and variations

1. Warm-up · Flash $f(x) = 3x - 1$. Calculate $\dfrac{f(4) - f(1)}{4 - 1}$.

2. Core · Flash Direction of variation of $f(x) = -4x + 7$?

3. Core + · Proof Show that $f(x) = 5x - 2$ is increasing on $\mathbb{R}$.

4. Back · Table Ages of the pupils in a club.

Age	10	11	12	13
Count	3	6	10	6

How many pupils are at most 11 years old? What relative frequency?

5. Running thread · Program → formula Procedure: multiply by 3 then subtract 2. Give $f(x)$ then $f(4)$.

Sheet 46 Thu. 4 March 2027 Absolute value and square functions: sign and variations

1. Warm-up · Flash Calculate $|-5|$, $(-5)^2$ & $|2 - 7|$.

2. Core · Method Give the variation table of $x \mapsto |x|$ and its sign.

3. Core + · Reasoning Compare $(-3)^2$ & $(-2)^2$ then 3^2 and 2^2. Is the square function increasing on $\mathbb{R}$?

4. Back · Method Two fair dice are rolled. Probability that the sum is 7?

5. Running thread · Method Procedure: choose a number; add 1; square it; subtract the square of the starting number. What is the result for 6? What general expression?

Sheet 47 Mon. 8 March 2027 Reciprocal function: variations; relative position of y = x and y = x^2

1. Warm-up · Flash Calculate $\dfrac{1}{2}$, $\dfrac{1}{-4}$ and $\dfrac{1}{0{,}1}$.

2. Core · Method Compare $\dfrac{1}{3}$ and $\dfrac{1}{5}$; what can be said about the variations of the reciprocal function on $]0; +\infty[$?

3. Core + · True/false True or false? Justify. For every real number $x > 0$, $x^2 \geqslant x$.

4. Back · Flash What are the front view and the top view of a cylinder standing on its base?

5. Running thread · Prove Show that the product of two even integers is a multiple of 4.

Sheet 48 Tue. 9 March 2027 Compare f (a) and $f(b)$; optimisation problems

1. Warm-up · Flash $f(x) = x^2$. Compare $f(-3)$ and $f(2)$.

2. Core · Method $f(x) = x^2$, $a = -1.5$ and $b = -2$. Compare $f(a)$ and $f(b)$ without calculation, using the variations.

3. Core + · Optimisation $A(x) = x(10 - x)$ for $x \in [0; 10]$ (area of a rectangle with perimeter 20). Show that $A(x) = 25 - (x - 5)^2$ and deduce the maximum.

4. Back · Flash 18 out of 24 pupils have an average above 10. Give the relative frequency as a percentage.

5. Running thread · MCQ The solution of $\dfrac{x}{3} + 1 = \dfrac{7}{2}$ is:

A. $\dfrac{15}{2}$ **B.** $\dfrac{9}{2}$ **C.** $\dfrac{21}{2}$ **D.** $\dfrac{5}{6}$

Cross-tabulating two categorical variables

Sheet 49 Thu. 11 March 2027 Cross-tabulating two categorical variables

1. Warm-up · Flash Express $\frac{24}{60}$ as a percentage.
2. Core · Flash In a class of 30 pupils, 18 are girls. What is the frequency of girls? Of boys?
3. Core + · Classify Classify each variable: eye colour; baccalauréat grade (AB, B, TB); height in cm.
4. Back · Problem A rectangle with perimeter 20 cm has length x. Express its area $A(x)$; calculate $A(3)$ and $A(5)$.
5. Running thread · Flash Calculate the exact volume of a cylinder with radius 2 cm and height 5 cm.

Sheet 50 Mon. 15 March 2027 Qualitative variables; two-way frequency table

1. Warm-up · Flash Complete: $12 + \ldots = 30$ and $22 + \ldots = 40$.
2. Core · Table Total 40 students. Girls: 22; half-boarders (DP): 25; DP girls: 15. Complete the two-way table.

	DP	Non-boarders	Total
Girls	15	…	22
Boys	…	…	…
Total	25	…	40

3. Core + · True/false True or false? Justify. "Satisfaction level (low, medium, high)" is a nominal qualitative variable.
4. Back · Problem Discount A: -30% off 80,000 Ar. Discount B: -20% then -10%. Which is the better deal?
5. Running thread · Method We know that $AB^2 + AC^2 = BC^2$. What conclusion do we draw? What property?

Sheet 51 Tue. 16 March 2027 Marginal frequencies

1. Warm-up · Flash Express $\frac{25}{40}$ as a percentage.
2. Core · Method With the table from sheet 50: calculate the marginal frequency of girls and that of the half-boarders.
3. Core + · Flash The marginal frequency of girls is 55%. What is that of boys?
4. Back · Flash Calculate $(2a - b)^2$ for $a = 3$ and $b = -2$.
5. Running thread · Flash $A(-1;2)$ and $B(3;2)$. What can be said about the line (AB) ? Give AB.

Sheet 52 Thu. 18 March 2027 Conditional frequencies

1. Warm-up · Flash Give $\frac{15}{22}$ as a percentage rounded to one decimal place.
2. Core · Method With the same table: among the girls, what is the frequency of DP pupils? Among the boys?
3. Core + · Distinguish Distinguish "the frequency of girls among DP pupils" from "the frequency of DP pupils among the girls".
4. Back · Construction steps We draw $[AB]$ of 6 cm, then the circles with centres A and B and radius 4 cm, which intersect at C and D. Type of $ACBD$?
5. Running thread · Flash Is 91 prime?